A Federated Learning Framework for Fingerprinting-Based Indoor Localization in Multibuilding and Multifloor Environments

Bo Gao, *Member, IEEE*, Fan Yang, Nan Cui, Ke Xiong, *Member, IEEE*, Yang Lu, *Member, IEEE*, and Yuwei Wang

***Abstract*—The participatory nature of federated learning (FL) makes it attractive for fingerprinting-based indoor localization in multibuilding and multifloor environments. A group of sensing clients can collaboratively leverage their private, local fingerprint data to help their edge server update a location prediction model. However, it is challenging to jointly handle the two involved issues, i.e., building-floor classification (BFC) and latitude–longitude regression (LLR), in a wide 3-D space through enabling FL on decentralized yet heterogeneous data and over an imperfect wireless network. In this article, we confront these challenges and propose an FL framework, FedLoc3D, for both BFC and LLR. Specifically, the former issue is addressed by an FedDSC-BFC approach, which generates a multilabel classification model based on a convolutional neural network with depthwise separable convolutions. The latter issue is addressed by an FedADA-LLR approach, which develops a multitarget regression model based on a deep neural network with autoencoder and data augmentation. Extensive experiments on a real-world data set of WiFi fingerprints are carried out, and our approaches with enhanced capabilities of feature extraction, generalization, and convergence are validated to improve both localization accuracy and learning efficiency under data heterogeneity and network instability.**

***Index Terms*—Edge computing, indoor localization, Internet of Things, machine learning, mobile computing.**

I. Introduction

RECENT advances in artificial intelligence (AI), especially in deep learning (DL), have been driving new breakthroughs in indoor localization [1], [2], [3]. As one of the most promising data-driven techniques, fingerprinting-based indoor localization can be augmented by AI through training a DL model for location prediction on a data set of location-labeled radio frequency (RF) fingerprints [4]. The location prediction model trained on the fingerprint data, e.g., a deep neural network (DNN), convolutional neural network (CNN), or recurrent neural network (RNN), maps the measurements of any target's RF fingerprint to a location estimate. Unlike geometric techniques, e.g., trilateration or triangulation, such a data-driven localization technique can work without the need for carefully capturing time- or angle-of-arrival measurements as well as accurately locating access points or base stations.

Mostly, fingerprinting-based indoor localization is achieved via training a DL model centrally at one spot, but this centralized learning approach does not work well in a dynamic environment. First, it can be costly to retrain the location prediction model on a central entity. The implementation of the deep model training is usually not easy for a resource-constrained platform, e.g., an edge server. Furthermore, fingerprinting-based localization is highly sensitive to environmental changes, e.g., relocated furniture, walking people, and added or removed access points [5]. To keep adapting to such dynamic changes, the model has to be frequently retrained, thus leading to significant computational overhead. Second, it can be costly to update the fingerprint data on a central entity. The process of a site survey is generally time consuming and labor intensive [6]. Moreover, large volumes of fresh training data may be repeatedly collected by one or more sensing devices due to environmental changes, especially in a wide space. To retrain the model in response to dynamic changes, the collected data have to be intensively uploaded over a bandwidth-limited system, e.g., a wireless network, thus generating expensive communication overhead. Third, it can be privacy invasive to share the locally collected fingerprint data with a central entity. Though crowdsourcing fresh training data from contributing devices is often useful in sufficiently capturing environmental changes, especially in a wide space [7], location-labeled RF fingerprints can easily reveal where the contributors are located or have visited. To guarantee overall localization performance in presence of dynamic changes, individual location privacy has to be totally sacrificed, thus resulting in extra privacy cost.

Federated learning (FL) is an emerging paradigm that can enable contributing devices to collaboratively train a DL model without sharing their private data [8], [9], [10], [11], [12], [13], so it perfectly meets the needs of fingerprinting-based localization. Specifically, one edge server and multiple sensing clients

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(various mobile or embedded devices capable of measuring RF fingerprints) form a wireless system for FL-empowered localization, in which the server globally builds a location prediction model that serves a whole indoor space with the help of the clients who locally own private fingerprint data and on-device computing resources. The server's global model is an aggregation of the clients' local models that are trained on their local data. Such a distributed learning approach based on the fusion of on-device learning and federated collaboration can better adapt to a dynamic environment compared with the centralized counterpart, because the use of FL brings manifold benefits, such as richer sensing/learning resources from contributing clients, faster direct responses to new measurements, better controllable overhead for model uploading, and lower privacy loss without data sharing. Furthermore, the participatory nature of FL makes it well applicable to fingerprinting-based localization in a sizable space, e.g., a building complex, in which contributing clients can be widely dispersed. Thanks to these benefits, FL is expected to become a new enabler of indoor localization, and some initial attempts have been made toward FL-empowered localization [14], [15], [16], [17], [18], [19], [20]. Except for the only work [20] that has considered a 3-D space, nearly all these existing studies have tried to implement FL for the development of distributed learning approaches to indoor localization on a relatively small 2-D plane, i.e., on a certain floor in a certain building. However, the trajectories of sensing clients are usually not confined to a single building or floor, so their local data with widely distributed 3-D location labels should have been fully crowdsourced and utilized to create a universally accurate global model that can serve multiple buildings and floors. As for [20], though a multibuilding and multifloor environment has been paid attention to, the prediction of building-floor-location information has been approximated as a classification task that can merely generate discretized, coarse-grained location estimates. The power of FL should have been better exploited to offer fine-grained localization services.

Despite recent efforts, it is still challenging to apply the notion of FL to fingerprinting-based localization, especially in a multibuilding and multifloor environment. First, locating any target in a wide 3-D indoor space requires to train a global model that makes both discrete building-floor estimates and continuous latitude–longitude estimates [21]. Clearly, this localization problem has to involve two different types of machine learning subproblems—building-floor classification (BFC) and latitude–longitude regression (LLR), which should be jointly solved. Second, establishing a global model through combining local models from contributing clients requires to deal with the heterogeneity of their local resources, particularly nonindependent and identically distributed (non-IID) data [22], [23]. Different clients can collect different amounts of local training data in different areas of interest using different devices, so the collected data are usually non-IID. It is highly likely that no or few clients need to visit every corner of their large 3-D space. If without guidance or incentive from the server, every client may have a distinct list of unvisited reference points or data collection points, resulting in local data sets with missing location labels, i.e., a non-IID setting of high degree [24]. This may generate location-biased local models [14]. It is important to generalize a universally accurate global model on those non-IID data. Third, enabling FL over a wireless network requires to optimize learning efficiency as well as localization accuracy under the instability of system collaboration. Imperfect wireless communications between collaborating devices may lead to bottlenecks in their federated model training [23], possibly causing the issues of straggling clients, unreachable training data, or even failed model aggregation. Although it is prioritized to achieve maximal localization accuracy, it is also inevitable to ensure sufficiently high learning efficiency to tolerate or circumvent such wireless bottlenecks.

In this article, we tackle the challenges above and make the contributions as follows.

1) We propose an FL framework, named FedLoc3D, for fingerprinting-based indoor localization in multibuilding and multifloor environments. It allows an edge server and multiple sensing clients to collaboratively train a location prediction model on decentralized fingerprint data. This framework is developed indeed for both BFC and LLR, and these two subproblems are decoupled and separately solved.
2) We propose an FedDSC-BFC approach, which addresses BFC via centralized pretraining and federated training of a multilabel classification model based on a 1-D CNN with depthwise separable convolutions. The implementations of centralized pretraining, autoencoder, and convolution operations help enhance its capabilities of feature extraction, generalization, and convergence for BFC.
3) We propose an FedADA-LLR approach, which addresses LLR via centralized pretraining and federated training of a multitarget regression model based on a DNN with autoencoder and data augmentation. The implementations of centralized pretraining, autoencoder, and data collaborations help enhance its capabilities of feature extraction, generalization, and convergence for LLR.
4) We conduct extensive experiments on a real-world data set of WiFi fingerprints, and our approaches are validated to improve both localization accuracy and learning efficiency under data heterogeneity and network instability.

The remainder of this article is organized as follows. Related work is discussed in Section II. Our FL framework for indoor localization is outlined in Section III, where the localization problem is formulated and further decoupled into the two subproblems. Our approaches to the subproblems of BFC and LLR are elaborated in Sections IV and V, respectively. Performance evaluation is presented in Section VI. Finally, our conclusion is summarized in Section VII.

II. Related Work

Ever since the success of DL, there have been recent studies that explore centralized learning approaches to fingerprinting-based indoor localization. Different types of DL models have

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been applied to map the features of any RF fingerprint to a location estimate, e.g., DNNs for reliable feature extraction from a large set of noisy input signals [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], CNNs for improved complexity in model training and enhanced robustness to input fluctuations [33], [35], [36], [37], [38], and RNNs for motion trajectory prediction from a sequence of correlated input signals [39], [40], [41], [42]. The majority of the above-mentioned studies have focused on the issue of LLR on a 2-D plane, except some work [27], [36] addressing only the issue of BFC and some other work [29], [30], [33], [34], [37] handling both the issues of BFC and LLR in a wide 3-D space. However, all these centralized learning approaches have become a second choice if their RF environment is truly dynamic. Because if so, the sensing clients have to constantly upload volumes of possibly private raw data and the edge server has to frequently retrain its DL model using limited computing resources of its own. Fortunately, the advent of FL has brought manifold benefits to fingerprinting-based localization in a dynamic environment, such as richer sensing/learning resources from contributing clients, faster direct responses to new measurements, better controllable overhead for model uploading, and lower privacy loss without data sharing.

Owing to the benefits of FL, increasing attention has been devoted to the development of distributed learning approaches to fingerprinting-based indoor localization. Some initial attempts have been made to enable FL-empowered localization [14], [15], [16], [17], [18], [19], [20]. Specifically in [14], a typical FL framework for LLR is proposed, in which a DNN model with deep autoencoder is collaboratively trained on decentralized, constantly updated fingerprints and thus is dynamically adaptable to environmental changes. In [15], a similar FL framework for LLR is introduced, in which a DNN model but without autoencoder is globally built through aggregating crowdsourced, locally derived such models. In [16], a general FL framework for LLR is presented, and use cases based on either a DNN model or a Gaussian process regression (GPR) model are studied where the model for target tracking is collectively learned on decentralized mobile trajectories. In [17], a personalized FL framework for LLR is designed, in which the training of a multilayer perceptron (MLP) model relies on the use of online semi-supervised learning that allows possibly moving clients to annotate newly collected unlabeled fingerprints with pseudo labels and at the same time train their local models on those pseudo-labeled fingerprints, with the aim of offering personalized localization services. In [18], another personalized FL framework for LLR is developed, in which a graph neural network (GNN) model is combined with self-attention to extract spatial features from fingerprint inputs and learn the ways of aggregating local embeddings or models toward personalized outcomes. In [19], uncertainty in the localization outputs of a probabilistic FL framework for LLR is addressed, and an approximation to a Bayesian neural network (BNN) model is achieved by applying Monte Carlo dropout so that more reliable local models can be identified for a more accurate global model. In [20], a pseudo-label-driven FL framework for building-floor-location classification (i.e., BFC and discretized LLR) is devised, in which a multilabel classifier with stacked autoencoder is first pretrained on labeled fingerprints to annotate a large volume of unlabeled samples with pseudo labels and the model is then cooperatively retrained on both labeled and pseudo-labeled fingerprints. However, nearly all these distributed learning approaches have dealt with only the issue of LLR on a relatively small 2-D plane, except the closest work [20] that has been interested in both the issues of BFC and LLR in a 3-D space but the latter regression task supposed to specify continuous latitude–longitude values has been simplified to a classification task merely able to generate discretized, coarse-grained location estimates. None of them has leveraged FL like our work to jointly handle both BFC and LLR in a wide 3-D space, which are different in essence. Besides, the above-mentioned studies have not worked out an FL framework like ours that is capable of balancing localization accuracy with learning efficiency under data heterogeneity and network instability.

III. Federated Learning Framework

In this section, we describe the system model that underpins our FL framework, FedLoc3D, for fingerprinting-based indoor localization in multibuilding and multifloor environments.

A. Fingerprinting-Based Localization

Any target user is expected to locate itself and request location-based services in a sizable 3-D indoor space, e.g., a building complex, where there are multiple buildings, each with multiple floors. To this end, a fingerprinting-based localization technique can be employed, taking advantage of the measurements of RF signals covering the target space, e.g., widely available WiFi fingerprints.

Most often, a DL model for location prediction is centrally trained on an edge server. Suppose that a data set of location-labeled RF fingerprints, say $\mathcal{D} = \{\boldsymbol{x}_i, \boldsymbol{y}_i\}|_{i\in\mathcal{N}}$, is available for deep model training, where $\mathcal{N} = \{1, 2, \ldots, N\}$ is the set of data sample indexes. Each RF fingerprint i consists of $\boldsymbol{x}_i$, a vector of RF measurements, and $\boldsymbol{y}_i$, a set of location label elements. Taking the UJIIndoorLoc data set [21] as an example, we consider a form of fingerprinting based on received signal strength (RSS) measurements, but our work is still extendible to accept other types of RF measurements. Then, for any $\boldsymbol{x}_i = (\mathrm{RSS}_{i,j})|_{j\in\mathcal{M}} \in \mathcal{D}$, on the one hand, a vector of M RSS values has been locally observed at a reference point covered by a group of access points and $\mathcal{M} = \{1, 2, \ldots, M\}$ is the set of access point indexes, where each $\mathrm{RSS}_{i,j}$ links to an access point j. For the corresponding $\boldsymbol{y}_i = (\mathrm{BLDG}_i, \mathrm{FLR}_i, \mathrm{LAT}_i, \mathrm{LONG}_i) \in \mathcal{D}$, on the other hand, a location label has been precisely specified for the reference point, where the former pair of $(\mathrm{BLDG}_i, \mathrm{FLR}_i)$ are discrete building-floor indexes and the latter pair of $(\mathrm{LAT}_i, \mathrm{LONG}_i)$ are continuous latitude–longitude values. Further procedures of data sample preprocessing will be suggested later.

The objective of fingerprinting-based localization is to learn a location prediction model, say $f(\cdot; \boldsymbol{\omega})$ with parameters $\boldsymbol{\omega}$, that

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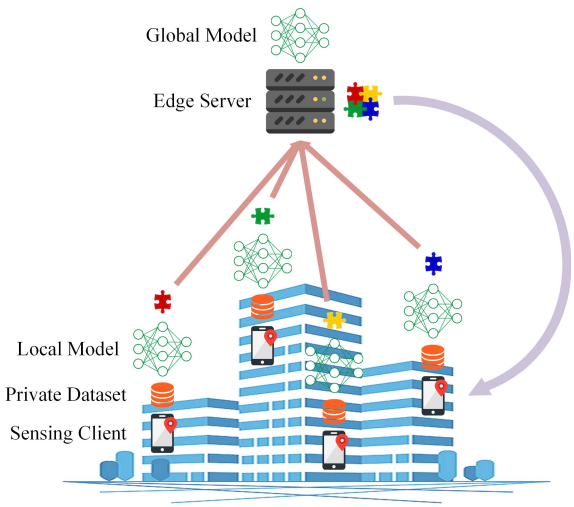

Fig. 1. Basic idea of FL-empowered localization.

minimizes an average loss function $\varepsilon(\boldsymbol{\omega})$, namely

$$\min_{\boldsymbol{\omega}\in\mathbb{R}^d} \varepsilon(\boldsymbol{\omega}). \tag{1}$$

Given the data set $\mathcal{D}$ of N data samples, we can define

$$\varepsilon(\boldsymbol{\omega}) \triangleq \frac{1}{N}\sum_{i\in\mathcal{N}} \ell\big(\boldsymbol{x}_i, \boldsymbol{y}_i; \boldsymbol{\omega}\big) \tag{2}$$

where $\ell(\boldsymbol{x}_i, \boldsymbol{y}_i; \boldsymbol{\omega})$ is a loss function that characterizes the error between ith location estimate $\hat{\boldsymbol{y}}_i = f(\boldsymbol{x}_i; \boldsymbol{\omega})$ and its ground truth $\boldsymbol{y}_i$. The definitions of ℓ will be discussed later.

B. FL-Empowered Localization

Following the principle of FL, a DL model for location prediction can be collaboratively trained over a wireless system consisting of an edge server and multiple sensing clients, as illustrated in Fig. 1. Suppose that each client $k \in \mathcal{K}$ locally owns a private data set of location-labeled RF fingerprints, say $\mathcal{D}_k = \{\boldsymbol{x}_i, \boldsymbol{y}_i\}|_{i\in\mathcal{N}_k}$, for federated model training, where $\mathcal{K} = \{1, 2, \ldots, K\}$ is the set of sensing client indexes and $\mathcal{N}_k = \{1, 2, \ldots, N_k\}$ is the set of data sample indexes. Furthermore, assume that $\mathcal{D} = \cup_{k\in\mathcal{K}}\mathcal{D}_k$.

The objective of FL-empowered localization is to learn a global model, say $f(\cdot; \bar{\boldsymbol{\omega}})$ with parameters $\bar{\boldsymbol{\omega}}$, that minimizes an average loss function $\varepsilon(\bar{\boldsymbol{\omega}})$. Because there are K contributing clients who collaboratively participate in the minimization of ε, we can rewrite the objective in (1) as follows:

$$\min_{\bar{\boldsymbol{\omega}}\in\mathbb{R}^d} \varepsilon(\bar{\boldsymbol{\omega}}) = \sum_{k\in\mathcal{K}} \frac{N_k}{N}\varepsilon_k(\bar{\boldsymbol{\omega}}) \tag{3}$$

where each $\varepsilon_k(\cdot)$ is an average loss function on $\mathcal{D}_k$. For each client k who is responsible for training a local model on the data set $\mathcal{D}_k$ of N_k data samples, we can define the following:

$$\varepsilon_k(\bar{\boldsymbol{\omega}}) \triangleq \frac{1}{N_k}\sum_{i\in\mathcal{N}_k} \ell\big(\boldsymbol{x}_i, \boldsymbol{y}_i; \bar{\boldsymbol{\omega}}\big) \text{ for } k \in \mathcal{K}. \tag{4}$$

Generally, the minimization of ε can be achieved through the minimization of ε_k for $k \in \mathcal{K}$.

A possible solution to the FL-empowered localization problem is to iteratively refine the global and local models. Upon receiving the last global model $\bar{\boldsymbol{\omega}}^t$ from the server, in iteration $t+1$, every client k updates its local model $\boldsymbol{\omega}_k^{t+1}$, so that

$$\min_{\boldsymbol{\omega}_k^{t+1}\in\mathbb{R}^d} \varepsilon_k\Big(\boldsymbol{\omega}_k^{t+1}\Big) \text{ for } k \in \mathcal{K}. \tag{5}$$

After collecting those local models from the clients, the server then aggregates them into its global model $\bar{\boldsymbol{\omega}}^{t+1}$, e.g., through

$$\bar{\boldsymbol{\omega}}^{t+1} = \sum_{k\in\mathcal{K}} \frac{N_k}{N}\boldsymbol{\omega}_k^{t+1}. \tag{6}$$

Our FL framework for indoor localization extends the regular solution concept with a focus on new workflows and model designs tailored to the needs of FL-empowered localization in a wide 3-D space, and our framework is compatible with not only the above FedAvg algorithm [8] but also other variants like the FedProx algorithm [43].

C. Subproblems of BFC and LLR

In a multibuilding and multifloor environment, the location prediction model is required to output both discrete building-floor estimates and continuous latitude–longitude estimates. However, this localization problem involves two different types of machine learning subproblems—BFC and LLR, which cannot be fully addressed by a single DL model at a time.

In our FedLoc3D framework, the subproblems of BFC and LLR are both solved via federated collaboration, but two separate DL models are learned. Specifically, the original data set $\mathcal{D}$ (including each client k's data set $\mathcal{D}_k$ for $k \in \mathcal{K}$) is used as two data sets: 1) $\mathcal{D}^C = \{\boldsymbol{x}_i, \boldsymbol{y}_i^C\}|_{i\in\mathcal{N}}$ (including $\mathcal{D}_k^C$ for $k \in \mathcal{K}$) for BFC, where each $\boldsymbol{y}_i^C = (\text{BLDG}_i, \text{FLR}_i) \in \mathcal{D}^C$ and 2) $\mathcal{D}^R = \{\boldsymbol{x}_i, \boldsymbol{y}_i^R\}|_{i\in\mathcal{N}}$ (including $\mathcal{D}_k^R$ for $k \in \mathcal{K}$) for LLR, where each $\boldsymbol{y}_i^R = (\text{LAT}_i, \text{LONG}_i) \in \mathcal{D}^R$. Then, a global classification model for BFC, say $f^C(\cdot; \bar{\boldsymbol{\omega}}^C)$, is collectively trained on local data sets $\mathcal{D}_k^C$ for $k \in \mathcal{K}$, and a global regression model for LLR, say $f^R(\cdot; \bar{\boldsymbol{\omega}}^R)$, is collectively trained on local data sets $\mathcal{D}_k^R$ for $k \in \mathcal{K}$. In the end, a combination of these two models can still generate an output of $\hat{\boldsymbol{y}}_i = (\hat{\boldsymbol{y}}_i^C, \hat{\boldsymbol{y}}_i^R) = (f^C(\boldsymbol{x}_i; \bar{\boldsymbol{\omega}}^C), f^R(\boldsymbol{x}_i; \bar{\boldsymbol{\omega}}^R))$ given an input of $\boldsymbol{x}_i$.

D. FedLoc3D Framework

Our FL framework for indoor localization, as shown in Fig. 2, brings a collection of new features and enhancements to the workflows and model designs of FL-empowered localization. Specifically, a lightweight autoencoder is first utilized as the input layer to downsize any input vector of RF measurements for the purpose of data dimensionality reduction and feature extraction, since an input vector collected from a sizable 3-D space covered by many access points can be usually long and sparse. Then, two DL models, one for BFC and the other for LLR, work in parallel to convert each encoded input vector to a set of location label elements. The former model named FedDSC-BFC is a multilabel classification model based on a 1-D CNN with depthwise separable convolutions, whereas the latter model named FedADA-LLR is

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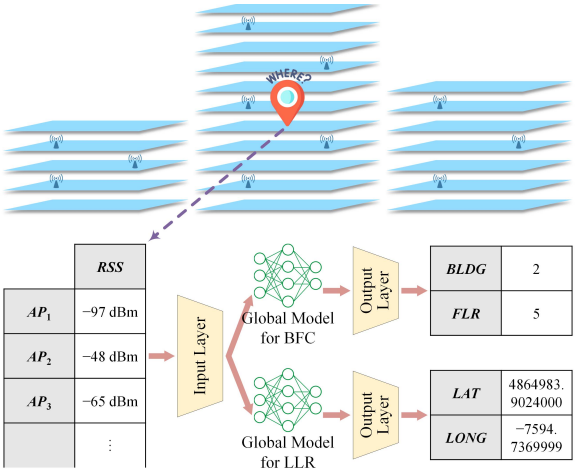

Fig. 2. Model components of the FedLoc3D framework.

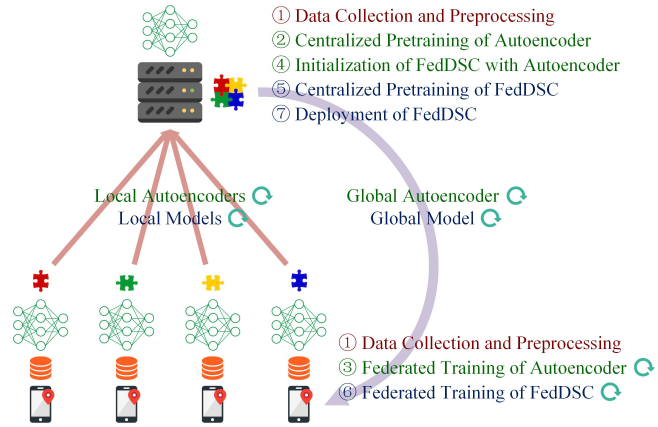

Fig. 3. Workflow of FedDSC-BFC.

a multitarget regression model based on a DNN with autoencoder and data augmentation. Given the same input vector, the output layer of the FedDSC-BFC produces a label of discrete building-floor indexes while that of the FedADA-LLR creates a label of continuous latitude–longitude values. All these model components are collaboratively trained via centralized pretraining at the server and federated training across the clients. Besides those private data sets locally collected by the clients, in our FL framework, a small central data set for model pretraining and data augmentation is globally managed by the server. This central data set can either be directly prepared by the server or be donated by some clients who trust the server to be privacy-protective. The above process of collaborative model training relies mainly on the clients' local data and additionally on the server's global data (including pseudo-generated data to be shared for local data augmentation). Our FedLoc3D framework indeed for both BFC and LLR aims to operate with enhanced capabilities of feature extraction and generalization and thus balance localization accuracy with learning efficiency under data heterogeneity and network instability. Our solutions to the subproblems of BFC and LLR will be elaborated in the next two sections, respectively.

IV. Building-Floor Classification

In this section, the first subproblem of BFC is addressed by an FedDSC-BFC approach, i.e., federated CNN with depthwise separable convolutions for BFC, that collaboratively learns a multilabel classification model to predict a target user's discrete building-floor indexes.

A. Workflow of FedDSC-BFC

As illustrated in Fig. 3, our FedDSC-BFC approach follows a workflow that involves a number of steps for the collaborative training of the classification model. In addition to a regular step for federated training, our approach incorporates a step for data preprocessing to mitigate the fluctuations of input signals. Furthermore, our approach takes two steps for centralized pretraining at the server to increase the convergence rate of federated training across the clients by uniting them with a common starting basis, so that improved learning efficiency can further help cut the involved computational and communication overhead. If possible, centralized pretraining on IID data at the server can also enhance the generalization outcome of federated training on non-IID data across the clients by indirectly mitigating their data or model biases, and thus can help achieve improved localization accuracy.

1) Data Collection and Preprocessing: Initially, the edge server prepares a small central data set $\mathcal{D}_0 = \{\boldsymbol{x}_i, \boldsymbol{y}_i\}|_{i\in\mathcal{N}_0}$ for centralized model pretraining, while each sensing client $k \in \mathcal{K}$ collects a private data set $\mathcal{D}_k = \{\boldsymbol{x}_i, \boldsymbol{y}_i\}|_{i\in\mathcal{N}_k}$ for federated model training. Compared with the locally collected $\mathcal{D}_k$ for $k \in \mathcal{K}$, the globally managed $\mathcal{D}_0$ can be a selection of RF fingerprints from a more complete set of reference points and can be made closer to IID. Each record of RF fingerprint i for BFC includes a vector $\boldsymbol{x}_i$ of RSS measurements along with a building-floor label $\boldsymbol{y}_i^C$ (a part of $\boldsymbol{y}_i$).

To mitigate the common issue of input signal fluctuations in fingerprinting-based localization, each $\mathrm{RSS}_{i,j} \in \mathcal{D}$ is transformed to a power representation of it [44], say

$$\mathrm{RSS}'_{i,j} = \begin{cases} \left(\frac{\mathrm{RSS}_{i,j}-\mathrm{MIN}}{-\mathrm{MIN}}\right)^{\alpha}, & \text{if detectable} \\ 0, & \text{if undetectable} \end{cases} \tag{7}$$

where MIN is the globally minimal RSS value, which can be easily obtained through message exchanges between the server and the clients, and $\alpha > 1$ is a constant. In the UJIIndoorLoc data set, for example, $\mathrm{RSS}_{i,j} \in [\mathrm{MIN} = -104, 0]$ dBm, so this transformation gives $\mathrm{RSS}'_{i,j} \in [0, 1]$. If an $\mathrm{RSS}_{i,j}$ is too weak to detect, manually set $\mathrm{RSS}'_{i,j} = 0$. As in Fig. 4, this power representation with a larger value of α places greater focus on the features of stronger signals with fewer fluctuations than those of weaker signals with more fluctuations. We assume that $\alpha = e$ (Euler's number).

2) Centralized Pretraining of Autoencoder: Whenever the data sets are ready for use, the process of training a lightweight autoencoder is initialized to support data dimensionality reduction and feature extraction, which includes a short initial step for centralized pretraining at the server and a short iterative step for federated training across the clients.

The server pretrains the autoencoder on the central data set $\mathcal{D}_0$ in a regular way. The objective of this task is to learn

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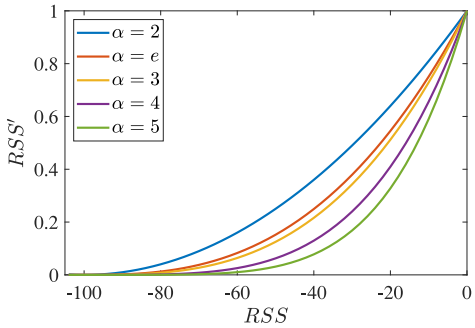

Fig. 4. Transformation of RSS to RSS′.

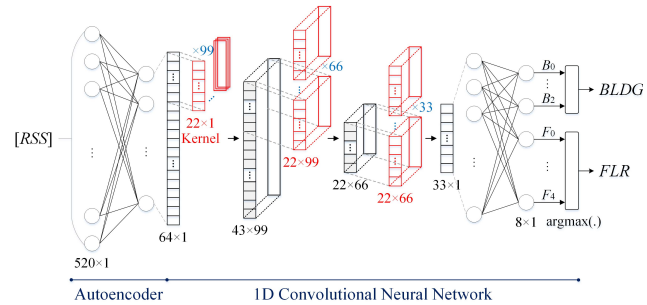

Fig. 5. Architecture of FedCNN-BFC.

a pair of encoder and decoder, say $g(\cdot;\boldsymbol{\phi})$ and $h(\cdot;\boldsymbol{\psi})$ with parameters $\boldsymbol{\phi}$ and $\boldsymbol{\psi}$, respectively, that minimizes an average reconstruction error function $\epsilon(\boldsymbol{\phi},\boldsymbol{\psi})$, namely

$$\min_{\boldsymbol{\phi},\boldsymbol{\psi}\in\mathbb{R}^d} \epsilon(\boldsymbol{\phi},\boldsymbol{\psi}). \tag{8}$$

Given the data set $\mathcal{D}_0$ of N_0 data samples, we can define

$$\epsilon(\boldsymbol{\phi},\boldsymbol{\psi}) \triangleq \frac{1}{N_0}\sum_{i\in\mathcal{N}_0} \ell(\boldsymbol{x}_i;\boldsymbol{\phi},\boldsymbol{\psi}) \tag{9}$$

where $\ell(\boldsymbol{x}_i;\boldsymbol{\phi},\boldsymbol{\psi})$ is a loss function that characterizes the error between ith reconstruction estimate $\hat{\boldsymbol{x}}_i = h(g(\boldsymbol{x}_i;\boldsymbol{\phi});\boldsymbol{\psi})$ and its ground truth $\boldsymbol{x}_i$. Only a small number of epochs are required to well train this basic model, without generating much additional overhead. At the end of this step, the server shares the resulting autoencoder, denoted by $(\bar{\boldsymbol{\phi}}^0, \bar{\boldsymbol{\psi}}^0)$, with the clients for further enhancement.

3) Federated Training of Autoencoder: Upon receiving the pretrained model, every client k begins to iteratively train a local autoencoder on its private data set $\mathcal{D}_k$. Given the latest global model $(\bar{\boldsymbol{\phi}}^t, \bar{\boldsymbol{\psi}}^t)$ from the server, in iteration $t+1$, every client k updates its local model $(\boldsymbol{\phi}_k^{t+1}, \boldsymbol{\psi}_k^{t+1})$, so that

$$\min_{\boldsymbol{\phi}_k^{t+1},\boldsymbol{\psi}_k^{t+1}\in\mathbb{R}^d} \epsilon_k\left(\boldsymbol{\phi}_k^{t+1},\boldsymbol{\psi}_k^{t+1}\right) \text{ for } k\in\mathcal{K} \tag{10}$$

where each $\epsilon_k(\cdot,\cdot)$ is an average reconstruction error function on $\mathcal{D}_k$. Given the data set $\mathcal{D}_k$ of N_k data samples, we can define ϵ_k similar to ϵ in (9). After collecting those local models from the clients, the server then aggregates them into its global model $(\bar{\boldsymbol{\phi}}^{t+1}, \bar{\boldsymbol{\psi}}^{t+1})$, e.g., through

$$\bar{\boldsymbol{\phi}}^{t+1} = \sum_{k\in\mathcal{K}} \frac{N_k}{N}\boldsymbol{\phi}_k^{t+1} \quad\text{and}\quad \bar{\boldsymbol{\psi}}^{t+1} = \sum_{k\in\mathcal{K}} \frac{N_k}{N}\boldsymbol{\psi}_k^{t+1}. \tag{11}$$

In the last iteration, the server can obtain the desired global autoencoder, denoted by $(\bar{\boldsymbol{\phi}},\bar{\boldsymbol{\psi}})$. Again, only a small number of iterations are required to well train it.

4) Initialization of FedDSC With Autoencoder: Once the process of autoencoder training is complete, the server takes the resulting encoder part $\bar{\boldsymbol{\phi}}$ to initialize the input layer of our multilabel classification model for BFC.

5) Centralized Pretraining of FedDSC: As the main task for BFC, the process of training the classification model, i.e., a 1-D CNN, includes an initial step for centralized pretraining at the server and an iterative step for federated training across the clients.

The server pretrains the CNN model on the central data set $\mathcal{D}_0$ in a regular way, as introduced in Section III-A. At the end of this step, in addition to distributing the autoencoder $\bar{\boldsymbol{\phi}}$, the server shares the resulting model, denoted by $\bar{\boldsymbol{\omega}}^{0,C}$, with the clients for further enhancement.

6) Federated Training of FedDSC: Upon receiving the pretrained model, every client k begins to iteratively train a local CNN model on its private data set $\mathcal{D}_k$ and contribute to federated collaboration led by the server, as introduced in Section III-B. Given the latest global model $\bar{\boldsymbol{\omega}}^{t,C}$ from the server, in iteration $t+1$, every client k updates its local model $\boldsymbol{\omega}_k^{t+1,C}$. In the last iteration, the server can finally obtain the desired global CNN model, denoted by $\bar{\boldsymbol{\omega}}^C$, which solves the first subproblem of BFC.

7) Deployment of FedDSC: Once the process of model training is complete, the server is able to offer a coarse-grained localization service, which predicts $\hat{\boldsymbol{y}}_i^C = f^C(\boldsymbol{x}_i;\bar{\boldsymbol{\phi}},\bar{\boldsymbol{\omega}}^C)$ based on any $\boldsymbol{x}_i$ using the newly learned $(\bar{\boldsymbol{\phi}},\bar{\boldsymbol{\omega}}^C)$.

B. Model Design of FedDSC-BFC

Our classification model for BFC takes a vector $\boldsymbol{x}_i$ of RSS measurements as an input. For a sizable 3-D space covered by a great number of access points, the length of an input vector can be very long. In the UJIIndoorLoc data set to be used by us, for example, each record of a WiFi fingerprint includes a vector of 520 RSS values that were measured somewhere in a 3-D space covered by 520 access points. Moreover, a typical input vector can be very sparse. Because of limited wireless coverage, the majority of the access points of interest should be undetectable at a certain location, resulting in many zero values in an input vector according to (7). To support data dimensionality reduction and feature extraction, as shown in Fig. 5, a fully connected autoencoder is applied in the input layer without generating much extra overhead. In this example, the autoencoder downsizes any input vector of length 520 to the one of length 64, as suggested by our experiments.

Given an encoded input vector, our model for BFC estimates a building-floor label $\hat{\boldsymbol{y}}_i^C$ as an output. In addition to the sparsity of input vectors, it is observed from data samples that the similarity of input vectors is high if they were collected on the same floor in the same building, i.e., with the same building-floor label, but is low otherwise. To facilitate feature extraction from such an input vector, as shown

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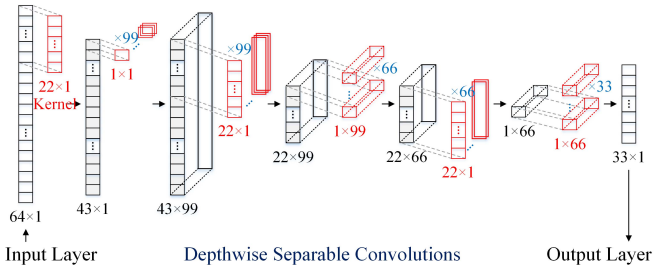

Fig. 6. Architecture of FedDSC-BFC.

in Fig. 5, we initially design an FedCNN-BFC approach, i.e., Federated CNN for BFC, in which 1-D convolution operations are performed a number of times before generating the output label. In this example, 99, 66, and 33 convolution kernels of sizes 22×1, 22×99, and 22×66 are applied to carry out convolution operations with stride 1, respectively. The convolutional layers convert the input vector of length 64 to a resulting vector of length 33. In the fully connected output layer, the desired building-floor label is determined. Here, in a target space consisting of three buildings, each with four to five floors, the output label is of length 8 bits as one-hot encoding is used, i.e., 3 bits for a building index and 5 bits for a floor index.

Noticing the large number of model parameters involved, however, we further develop a compressed variant of the CNN model, i.e., the FedDSC-BFC approach, to minimize computational and communication overhead. Efficient model compression can be achieved through performing depthwise separable convolutions [45] as in Fig. 6 instead of regular convolutions as in Fig. 5. In this example, 1, 99, 99, 66, 66, and 33 convolution kernels of sizes 22×1, 1×1, 22×1, 1×99, 22×1, and 1×66 are applied to carry out convolution operations with stride 1, respectively. As a result, nearly 80% of the regular model parameters can be removed.

V. Latitude–Longitude Regression

In this section, the second subproblem of LLR is addressed by an FedADA-LLR approach, i.e., federated DNN with autoencoder and data augmentation for LLR, that collaboratively learns a multitarget regression model to predict a target user's continuous latitude–longitude values.

A. Workflow of FedADA-LLR

As illustrated in Fig. 7, our FedADA-LLR approach follows a workflow that involves a number of steps for the collaborative training of the regression model. Particularly, our approach here incorporates two steps for data augmentation to enhance the generalization outcome of federated training on inadequate, non-IID data. It is usually time consuming and labor intensive for the clients to run a site survey, so their local data may be lacking for deep model training. Besides, the clients can collect different amounts of data in different areas of interest using different devices, so their local data may be non-IID. However, such inadequate, non-IID data are unlikely to well support FL-empowered localization, especially LLR that is more fine-grained than BFC. For example, the amount of fingerprint data

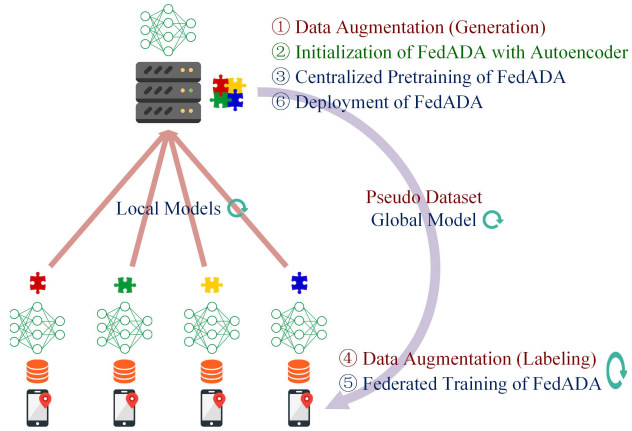

Fig. 7. Workflow of FedADA-LLR.

in the UJIIndoorLoc data set is relatively limited when being partitioned to simulate decentralized data sets, thus leading to a lack of high-quality data partitions for accurate localization, which happens a lot in practice. Together with centralized pretraining at the server, the augmentation of local data sets can further reinforce federated training across the clients, so that improved localization accuracy can be achieved.

1) Data Augmentation (Generation): Since the initial step for data collection, the edge server and the sensing clients have already obtained their own data sets $\mathcal{D}_0$ and $\mathcal{D}_k$ for $k \in \mathcal{K}$, respectively. Here, each record of RF fingerprint i for LLR includes a vector $\boldsymbol{x}_i$ of RSS measurements along with a latitude–longitude label $\boldsymbol{y}_i^R$ (a part of $\boldsymbol{y}_i$).

Instead of asking the clients to collect extra non-IID data samples [20], our approach cooperatively augments the local data sets through generating certain IID pseudo samples at the server while labeling and learning from them at the clients. In addition to facilitating the augmentation of inadequate, non-IID data, this design is beneficial in that it not only incentivizes the clients' participation by freeing them from additional measurement cost but also protects the clients' location privacy by preventing them from sharing any local data with others. In this former step for data augmentation, a pseudo data set of unlabeled RF fingerprints, say $\tilde{\mathcal{D}}^U = \{\tilde{\boldsymbol{x}}_i\}|_{i \in \tilde{\mathcal{N}}}$, is created at the server based on the central data set $\mathcal{D}_0$, where each $\tilde{\boldsymbol{x}}_i$ is an unlabeled RSS vector, and the pseudo data set is then distributed to the clients for further labeling in the latter step. The reason why the pseudo-generated $\tilde{\mathcal{D}}^U$ rather than the original $\mathcal{D}_0$ is shared with all the clients is that $\mathcal{D}_0$ may be donated by some clients who still care about their location privacy, but it would be not hard to infer those donators' location preferences if their real RF fingerprints are leaked. Our design has to guarantee the server to be privacy-protective and trustworthy enough.

Specifically through Algorithm 1, a pseudo data set $\tilde{\mathcal{D}}^U$ is created at the server based on the given data set $\mathcal{D}_0$, so that every client k's private data set $\mathcal{D}_k$ can be augmented by $\tilde{\mathcal{D}}^U$ (to be labeled). In this algorithm, unlabeled data samples are generated in groups, each of which is derived from a partition $\mathcal{D}_0^{\nu^C}$ of location-labeled data samples, considering the high similarity of data samples collected on the same floor in

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Algorithm 1 Data Augmentation

Input: location-labeled RF fingerprints in $\mathcal{D}_0$
Output: unlabeled RF fingerprints in $\tilde{\mathcal{D}}^U$

divide $\mathcal{D}_0$ into data partitions $\mathcal{D}_0^{\boldsymbol{y}^C}$ for all possible values of $\boldsymbol{y}^C$, i.e., all possible building-floor labels
for each building-floor label $\boldsymbol{y}^C$ **do**
  for each access point j **do**
    initialize $AVG_j^{\boldsymbol{y}^C}$, the average value of detectable $RSS_{i,j}$ for all data samples i in $\mathcal{D}_0^{\boldsymbol{y}^C}$, as undetectable
    if not all values of $RSS_{i,j}$ are undetectable **then**
      compute $AVG_j^{\boldsymbol{y}^C}$
    end if
    for each data sample i **do**
      generate $\widetilde{RSS}_{i,j}$, a pseudo value that is a random real number between $RSS_{i,j}$ and $AVG_j^{\boldsymbol{y}^C}$
    end for
  end for
end for
return $\tilde{\mathcal{D}}^U$, i.e., $\widetilde{RSS}_{i,j}$ for all values of i and j

the same building, i.e., with the same building-floor label $\boldsymbol{y}^C$. The data features of each $\boldsymbol{y}^C$ can be characterized by average RSS values $\mathrm{AVG}_j^{\boldsymbol{y}^C}$ for all access points j in $\mathcal{D}_0^{\boldsymbol{y}^C}$. For each actual measurement $\mathrm{RSS}_{i,j}$ in $\mathcal{D}_0$, a pseudo counterpart $\widetilde{\mathrm{RSS}}_{i,j}$ is generated, which is a random RSS value between $\mathrm{RSS}_{i,j}$ and $\mathrm{AVG}_j^{\boldsymbol{y}^C}$. In such a way without involving complex computations, each resulting $\tilde{\boldsymbol{x}}_i \in \tilde{\mathcal{D}}^U$ can approximate its corresponding $\boldsymbol{x}_i \in \mathcal{D}_0$, and the data sets $\tilde{\mathcal{D}}^U$ and $\mathcal{D}_0$ are of equal size.

2) Initialization of FedADA With Autoencoder: Both the DL models for BFC and LLR share the same autoencoder to avoid another process of autoencoder training, so the server directly takes the encoder part $\bar{\boldsymbol{\phi}}$ originally for BFC to initialize the input layer of our multitarget regression model for LLR.

3) Centralized Pretraining of FedADA: As the main task for LLR, the process of training the regression model, i.e., a DNN, includes an initial step for centralized pretraining at the server and two iterative steps for local data augmentation and federated training across the clients.

The server pretrains the DNN model on the central data set $\mathcal{D}_0$ in a regular way, as introduced in Section III-A. At the end of this step, the server shares the resulting model, denoted by $\bar{\boldsymbol{\omega}}^{0,R}$, with the clients for further enhancement.

4) Data Augmentation (Labeling): Upon receiving the pseudo data set as well as the pretrained model shared from the server, every client k starts to iteratively refine the labeling of unlabeled samples in the data set $\tilde{\mathcal{D}}^U$. Before each iteration, say $t+1$, of local data augmentation and federated training, client k has just received the latest global model $\bar{\boldsymbol{\omega}}^{t,R}$ (including the pretrained model $\bar{\boldsymbol{\omega}}^{0,R}$ when $t=0$) from the server. The idea of pseudo sample labeling is to utilize the previously trained model for LLR to directly annotate each unlabeled RSS vector $\tilde{\boldsymbol{x}}_i \in \tilde{\mathcal{D}}^U$ with a new location label $\tilde{\boldsymbol{y}}_i^{t+1,R}$. Thus, in this latter step for data augmentation, the pseudo data set is iteratively updated with refined pseudo labels at every client, and in iteration $t+1$, it becomes $\tilde{\mathcal{D}}^{t+1} = \{\tilde{\boldsymbol{x}}_i, \tilde{\boldsymbol{y}}_i^{t+1,R}\}|_{i\in\tilde{\mathcal{N}}}$. Then every client k's private data set $\mathcal{D}_k$ can be augmented by $\tilde{\mathcal{D}}^{t+1}$, and an integration of them gives the augmented data set $\tilde{\mathcal{D}}_k^{t+1}$ to be used for federated training. Taking advantage of the centrally generated $\tilde{\mathcal{D}}^{t+1}$, the resulting $\tilde{\mathcal{D}}_k^{t+1}$ for $k \in \mathcal{K}$ can be closer to IID than the original ones.

5) Federated Training of FedADA: Following the iterative step for local data augmentation, every client k aims to iteratively train a local DNN model on the augmented version of its private data set $\mathcal{D}_k$ and contribute to federated collaboration led by the server, as introduced in Section III-B. Before iteration $t+1$ of federated training, client k has obtained the latest global model $\bar{\boldsymbol{\omega}}^{t,R}$ as well as the current augmented data set $\tilde{\mathcal{D}}_k^{t+1}$. Then, every client k's local model $\boldsymbol{\omega}_k^{t+1,R}$ can be updated accordingly. In the last iteration, the server can finally build the desired global DNN model, denoted by $\bar{\boldsymbol{\omega}}^R$, which solves the second subproblem of LLR.

6) Deployment of FedADA: Once the process of model training is complete, the server is able to offer a fine-grained localization service, which predicts $\hat{\boldsymbol{y}}_i = (\hat{\boldsymbol{y}}_i^C, \hat{\boldsymbol{y}}_i^R) = (f^C(\boldsymbol{x}_i; \bar{\boldsymbol{\phi}}, \bar{\boldsymbol{\omega}}^C), f^R(\boldsymbol{x}_i; \bar{\boldsymbol{\phi}}, \bar{\boldsymbol{\omega}}^R))$ based on any $\boldsymbol{x}_i$ using the previously learned $(\bar{\boldsymbol{\phi}}, \bar{\boldsymbol{\omega}}^C)$ and newly learned $\bar{\boldsymbol{\omega}}^R$.

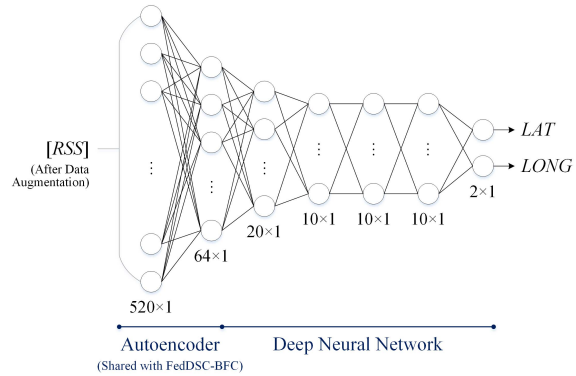

Fig. 8. Architecture of FedADA-LLR.

B. Model Design of FedADA-LLR

Our regression model for LLR takes a vector $\boldsymbol{x}_i$ or $\tilde{\boldsymbol{x}}_i$ of RSS measurements as an input. As above, an input vector collected from a sizable 3-D space covered by many access points can be usually long and sparse, so a lightweight autoencoder is still needed to effectively support data dimensionality reduction and feature extraction. To cut the cost for autoencoder training, as shown in Fig. 8, the same autoencoder is shared between the DL models for BFC and LLR in the input layer. In this example, the autoencoder downsizes any input vector of length 520 to the one of length 64.

Given an encoded input vector, our model for LLR estimates a latitude–longitude label $\hat{\boldsymbol{y}}_i^R$ as an output. Taking advantage of the augmented data and the autoencoder, as shown in Fig. 8, a regular fully connected DNN can sufficiently achieve an appropriate balance between localization accuracy and learning efficiency or model complexity. In this example, the hidden

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layers of sizes 20×1, 10×1, 10×1, and 10×1 convert the input vector of length 64 to a resulting vector of length 10. In the output layer, the desired latitude–longitude label is finally determined. The output label of length two real numbers given by the model for LLR further specifies a location estimate on the floor level suggested by the model for BFC.

VI. Performance Evaluation

In this section, we conduct extensive experiments to evaluate our FedLoc3D framework that addresses the issues of BFC and LLR under data heterogeneity and network instability.

A. Experimental Settings

Our experiments have been conducted on the most relevant and commonly used UJIIndoorLoc data set [21], which consists of 21 049 location-labeled WiFi fingerprints collected from three campus buildings, each with four to five floors. Each record of a WiFi fingerprint includes a vector of 520 RSS measurements along with a location label of four elements. The former two label elements are real-world longitude and latitude coordinates (in meters with UTM from WGS84), and the latter two label elements are floor and building indexes. These fingerprint samples were collected from a designated group of reference points. In any of the three buildings, the centroid point inside and the entrance point(s) outside every closed space (e.g., classroom, office, laboratory, and restroom) were chosen as reference points for data collection. As a result, there were total 924 reference points selected to cover a surface of 108 703 m^2, and 20 or 30 fingerprint samples were generated at each reference point.

This data set has been made applicable to FL by us. We have first randomly selected 20% of the original training data as our testing set, and we have given up using the original testing data that were potentially affected by environmental changes since the testing data were collected several months later than the training data. We have then grouped the remainder of the original training data into $K+1$ data partitions, i.e., $\mathcal{D}_0$ and $\mathcal{D}_k$ for $k \in \mathcal{K}$, as our training sets. The former partition is used to enhance FL, and has been created by randomly selecting 10% of the original training data. The latter K partitions are used to simulate FL on either IID or non-IID data. For the IID case, on the one hand, the data samples of each $\mathcal{D}_k$ have been randomly selected from the remaining data, so that the data distribution of each partition can be close to that of the original data set. For the non-IID case, on the other hand, the data samples of each $\mathcal{D}_k$ have been designedly selected by restricting each client k to certain preferred subareas of interest or subsets of reference points that have been manually taken from the target space, so that the data distributions of different partitions and that of the original data set can differ a lot. In fact, this way of data partitioning yields local data sets with largely missing location labels, i.e., a worst-case non-IID setting [24].

Our FedLoc3D framework has been implemented in the following settings. The autoencoder has been built using the rectified linear unit (ReLU) activation function and the mean squared error (MSE) loss function. It has been centrally pretrained for ten epochs and has further been collaboratively trained for ten iterations, each with ten epochs, both using a learning rate 0.001. Here, the notion of "epoch" is related to on-device learning and each epoch is referred to as one cycle of training either a pretrained model at the server or a local model at a client, while the notion of "iteration" is related to federated collaboration and each iteration is referred to as one round of uploading local models and aggregating them into a global model. The classification model for BFC has been built using the ReLU activation function and the binary cross entropy (BCE) loss function. It has been centrally pretrained for ten epochs and has further been collaboratively trained for 300 iterations, each with 20 epochs, both using a learning rate 0.0001 and a batch size 64. The regression model for LLR has been built using the ReLU activation function and the MSE loss function. It has been centrally pretrained for 700 epochs using a learning rate 0.001 and a batch size 64, and has further been collaboratively trained for 400 iterations, each with 20 epochs, using a learning rate 0.0002 and a batch size 64. In comparison, LLR is much more fine-grained than BFC, so it usually takes more iterations or epochs to build up a sufficiently accurate model for LLR than for BFC. Although it looks costly to pretrain the model for LLR, the involved epochs are just one-time cost and can also be kept to an acceptable level thanks to the small size of the central data set.

The evaluation of our FL framework has been performed while assuming the likelihood of unreliable network connections between the server and the clients, which can lead to the presence of straggling clients. In this situation, a certain percentage of such clients can be dropped to avoid overly low learning efficiency at some cost of localization accuracy [46]. For example, FL with a drop rate 20% is only supported by the 80% fastest clients out of the total K clients.

B. Evaluation of FedDSC-BFC

To validate the performance of our model for BFC based on 1-D CNN with depthwise separable convolutions, the FedDSC-BFC approach has been compared with a CenDNN-BFC approach [29], i.e., Centralized DNN for BFC, and two variants of our approach—an FedDNN-BFC approach, i.e., Federated DNN for BFC, and the FedCNN-BFC approach. The fully connected FedDNN-BFC consists of the same autoencoder that downsizes any input vector of length 520 to the one of length 64, one hidden layer of size 8×1 that converts the encoded input vector of length 64 to a resulting vector of length 8, and the output layer that generates the output label of length 8 bits.

The classification accuracy of our approaches for BFC has been examined, which is measured by the ratio of correct building-floor predictions to all predictions. The overall training performance of the FL approaches is compared in Fig. 9, when $K = 10$. It can be seen that for both the IID and non-IID cases, our FedCNN-BFC achieves the best results in both classification accuracy and convergence rate, and our FedDSC-BFC is generally comparable to it. Note that the FedDSC-BFC is a compressed variant of the FedCNN-BFC, and it will be shown that compared with the FedCNN-BFC with a very large-sized model, the FedDSC-BFC with

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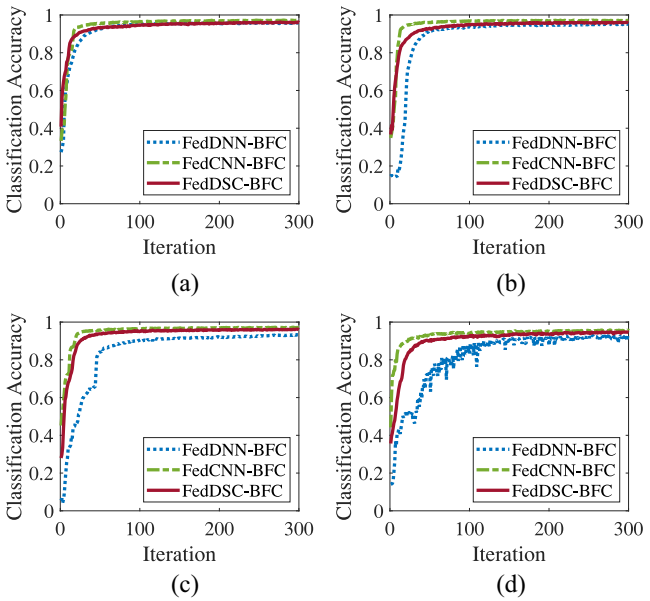

Fig. 9. Model training process for BFC. (a) IID data (drop rate 0%). (b) IID data (drop rate 20%). (c) Non-IID data (drop rate 0%). (d) Non-IID data (drop rate 20%).

a middle-sized model achieves better overhead performance without sacrificing much accuracy performance. Without our proposed features and enhancements, the FedDNN-BFC is the least efficient one especially on non-IID data that are particularly common in FL. The average accuracy performance of the classification approaches is compared in Fig. 10, and similar results can be observed. Compared with the CenDNN-BFC on centralized, IID data, our FedCNN-BFC, and FedDSC-BFC on decentralized, IID/non-IID data are just slightly worse in terms of classification accuracy. The FedDNN-BFC is the least accurate one without the conduct of convolution operations for enhanced feature extraction. In our framework, an autoencoder is shared between the DL models for BFC and LLR. The necessity of an autoencoder for BFC is assessed in Fig. 11, and that for LLR will be shown later. It can be seen that the removal of the autoencoder causes a certain degradation of accuracy performance, especially on non-IID data, so the autoencoder is useful in facilitating data dimensionality reduction and feature extraction without introducing much extra overhead. Furthermore, our framework is devoted to bringing new workflows and model designs of FL-empowered localization. Although model aggregation is not our primary focus and the most popular FedAvg algorithm is adopted by default, our framework is widely compatible with other variants like the well-known FedProx algorithm. The effectiveness of these algorithms in boosting the accuracy performance for BFC is estimated in Fig. 12. It can be seen that the FedProx offers a slight performance gain over the FedAvg while incurring additional computational overhead [47]. Here, we evaluate our approaches based on the FedAvg for minimal cost, but our framework is not limited to this setting.

The learning efficiency of our approaches for BFC has also been investigated, which is characterized by the ratio of involved communication overhead to reference overhead. Because the efficiency of FL over a wireless network largely relies on

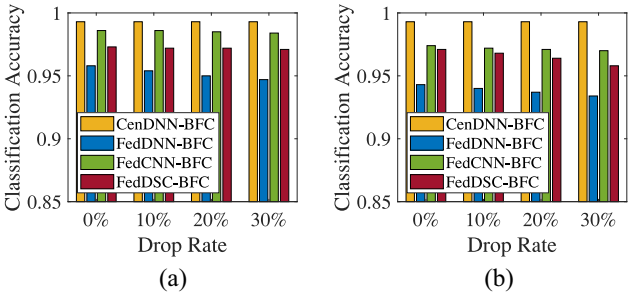

Fig. 10. Classification accuracy for BFC. (a) IID data. (b) Non-IID data.

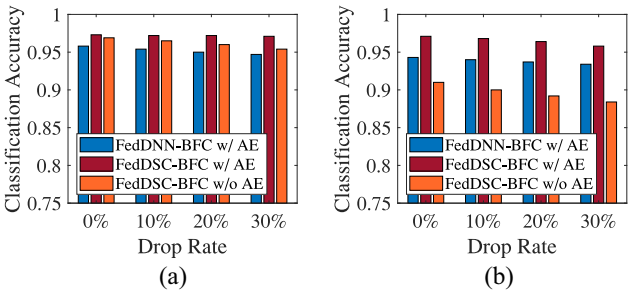

Fig. 11. Necessity of autoencoder for BFC. (a) IID data. (b) Non-IID data.

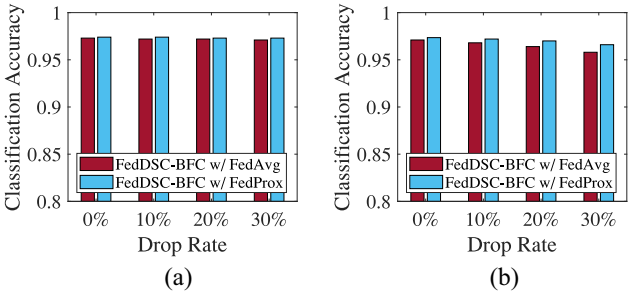

Fig. 12. Impact of model aggregation design for BFC. (a) IID data. (b) Non-IID data.

the timeliness of model exchanges between the server and the clients, we estimate the ratio of overhead through $\eta^C = ((|\bar{\boldsymbol{\omega}}^C| \times T)/(|\bar{\boldsymbol{\omega}}_0^C| \times T_0))$, where $|\bar{\boldsymbol{\omega}}^C|$ (or $|\bar{\boldsymbol{\omega}}_0^C|$) is the number of model parameters of an approach (or the FedDNN-BFC as a reference); and T (or T_0) is the number of training iterations of an approach (or the reference) that ensures a certain target accuracy. Here, the resulting level of accuracy performance achieved by the FedDNN-BFC as in Fig. 10 is used as the target accuracy. The average overhead performance of the FL approaches is compared in Fig. 13, given that the model sizes of FedDNN-BFC, FedCNN-BFC, and FedDSC-BFC are 33 936, 227 728, and 46 515, respectively, as in Table I. Although the FedDNN-BFC operates with a relatively small-sized model, our FedDSC-BFC generates less overhead with a middle-sized model due to faster convergence. In addition, the FedCNN-BFC is not always the most costly one despite exchanging a very large-sized model. This is because as shown in Fig. 9, our approaches can efficiently reach the target accuracy only in small numbers of iterations while the FedDNN-BFC mostly cannot. The ratio of overhead also changes with the drop rate if under network instability. As more clients are dropped, on the one hand, a larger number of iterations are normally

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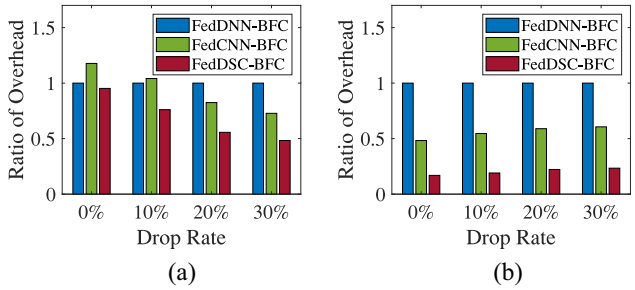

Fig. 13. Communication overhead for BFC. (a) IID data. (b) Non-IID data.

TABLE I
MODEL SIZES OF APPROACHES FOR BFC

Model	Number of Parameters
FedDNN-BFC	33,936
FedCNN-BFC	227,728
FedDSC-BFC	46,515

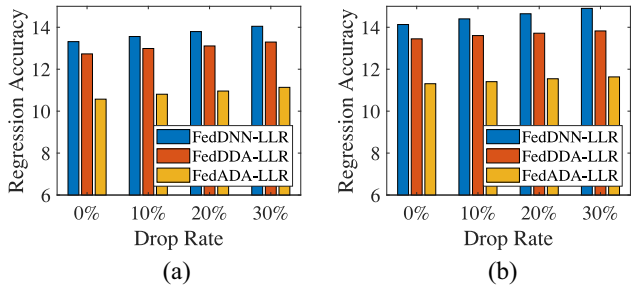

Fig. 14. Regression accuracy for LLR. (a) IID data. (b) Non-IID data.

needed for the training of an approach to maintain a certain level of accuracy performance. On the other hand, a lower target accuracy is probably given by the FedDNN-BFC as more clients become unavailable, and if so, a smaller number of iterations can usually be achieved for the training of an approach toward the reduced level of accuracy performance. Overall, the resulting values of T (and T_0) often get smaller with a higher drop rate since the impact of a lower target accuracy is more dominant. Compared with the decrease in $|\bar{\boldsymbol{\omega}}_0^C| \times T_0$ given by the FedDNN-BFC, the decrease in $|\bar{\boldsymbol{\omega}}^C| \times T$ given by our FedCNN-BFC or FedDSC-BFC is generally more significant, making ratios fall as in Fig. 13(a). Sometimes the accuracy requirement is greatly relaxed, e.g., on non-IID data and thus the values of T are small (in fact much smaller than T_0), but the decrease in $|\bar{\boldsymbol{\omega}}^C| \times T$ is relatively less significant, making ratios rise a little as in Fig. 13(b).

C. Evaluation of FedADA-LLR

To validate the performance of our model for LLR based on DNN with autoencoder and data augmentation, the FedADA-LLR approach has been compared with an FedDNN-LLR approach [15], i.e., Federated DNN for LLR, and a variant of our approach—an FedDDA-LLR approach, i.e., Federated DNN with only Data Augmentation for LLR. The FedDNN-LLR is also implemented according to the experimental settings as in Section VI-A.

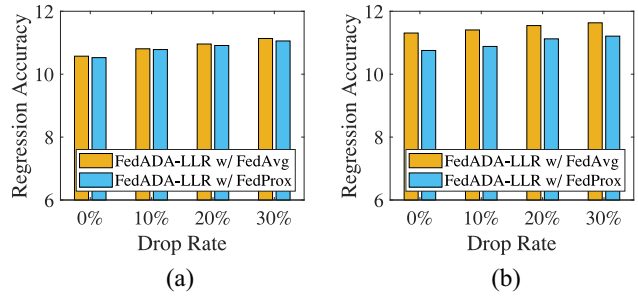

Fig. 15. Impact of model aggregation design for LLR. (a) IID data. (b) Non-IID data.

The regression accuracy of our approaches for LLR has been validated, which is measured by the error distance (in meters) between predicted location and ground truth on the 2-D plane. The average accuracy performance of the regression approaches is compared in Fig. 14, when the server generates 2000 pseudo samples and each client randomly receives 1200 out of them for data augmentation. It can be seen that for both the IID and non-IID cases, the integration of autoencoder and data augmentation makes our FedADA-LLR achieve the most accurate localization while the application of only data augmentation still helps our FedDDA-LLR reduce the error distance of the DNN model. Note that the resulting error distances here are relatively large, which are dependent on the data set and our way it is used. There are several concerns regarding the UJIIndoorLoc data set: 1) the reference points for data collection were sparsely distributed (i.e., 924 reference points covering a surface of 108 703 m^2, or one reference point per 117.6 m^2 on average); 2) a relatively limited number of diverse data samples were collected (i.e., sampling each reference point for two or three times, each time for ten similar measurements), especially when being cut down to prepare an extra testing set and being partitioned to simulate decentralized data sets; and 3) the data samples were collected by 18 mobile users with different heights using different devices, leading to inconsistent-quality measurements. Even so, our own experimental settings are still fair for all the approaches under comparison. Similar to Fig. 12, the effectiveness of model aggregation algorithms in boosting the accuracy performance for LLR is estimated in Fig. 15, and again, the FedProx offers a slight performance gain over the FedAvg while incurring additional computational overhead. Moreover, the effectiveness of our data augmentation algorithm depends on the number of pseudo samples each client is granted, as shown in Fig. 16. It can be observed that the provision of either too few or too many pseudo samples does not result in the best accuracy performance. One important reason is that the quality of augmented data may be harmed if the percentage of unlabeled samples with pseudo labels is too high, though it cannot be too low.

The learning efficiency of our approaches for LLR has also been studied, which is characterized by the ratio of involved communication overhead to reference overhead as above. Similarly, we estimate the ratio of overhead through $\eta^R = ((|\bar{\boldsymbol{\omega}}^R| \times T)/(|\bar{\boldsymbol{\omega}}_0^R| \times T_0))$, where $|\bar{\boldsymbol{\omega}}^R|$ (or $|\bar{\boldsymbol{\omega}}_0^R|$) is the number of model parameters of an approach (or the

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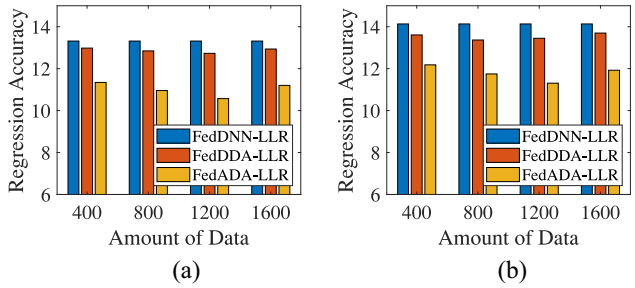

Fig. 16. Impact of pseudo data size for LLR. (a) IID data. (b) Non-IID data.

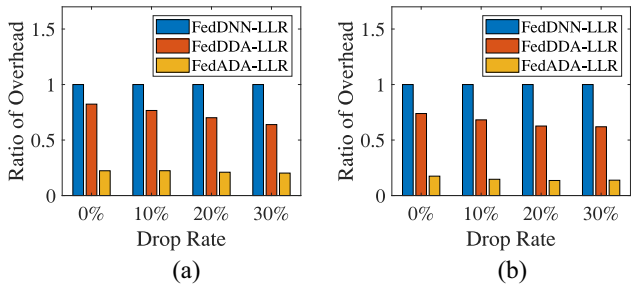

Fig. 17. Communication overhead for LLR. (a) IID data. (b) Non-IID data.

TABLE II
MODEL SIZES OF APPROACHES FOR LLR

Model	Number of Parameters
FedDNN-LLR	10,872
FedDDA-LLR	10,872
FedADA-LLR	35,096

FedDNN-LLR as a reference); and T (or T_0) is the number of training iterations of an approach (or the reference) that ensures a certain target accuracy. Here, the resulting level of accuracy performance achieved by the FedDNN-LLR as in Fig. 14 is used as the target accuracy. The average overhead performance of the FL approaches is compared in Fig. 17, given that the model sizes of FedDNN-LLR, FedDDA-LLR, and FedADA-LLR are 10 872, 10 872, and 35 096, respectively, as in Table II. Compared with the FedDNN-LLR on the original data, both our FedADA-LLR and FedDDA-LLR on the augmented data generate much less overhead for exchanging model parameters thanks to their faster convergence. Note that our approaches also involve extra one-time overhead for distributing a few pseudo samples to the clients, but it is generally negligible. Furthermore, the autoencoder is worth establishing, since it helps our FedADA-LLR most efficiently reach the target accuracy only in very small numbers of iterations.

D. Evaluation of FedLoc3D

Given each input vector of RSS measurements, a pair of output labels, i.e., a building-floor label and a latitude–longitude label, can be generated separately by the DL models for BFC and LLR. To validate the performance of our FedLoc3D framework as a whole, we need to unify any group of discrete

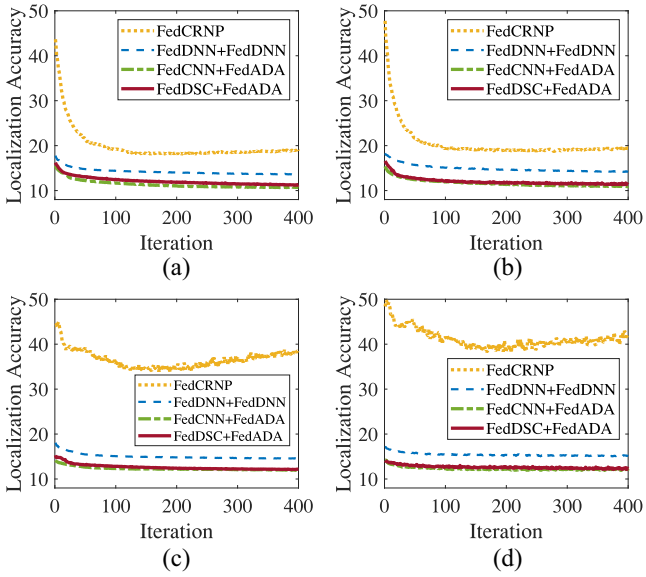

Fig. 18. Model training process for FedLoc3D. (a) IID data (drop rate 0%). (b) IID data (drop rate 20%). (c) Non-IID data (drop rate 0%). (d) Non-IID data (drop rate 20%).

building-floor indexes and continuous latitude–longitude values into a set of altitude, latitude, and longitude coordinates with the same unit. Because the altitude information is missing in the UJIIndoorLoc data set, we have to infer it from the floor information while assuming an average height of each floor level. Then for any location point, its value of altitude can be indicated by the model for BFC and its values of latitude and longitude can be given by the model for LLR. After unifying the unit of output labels, we have been able to compare different combinations of the above-evaluated approaches. Moreover, the closest work [20], named FedCRNP, has also been implemented by us as a benchmark, which approximates the prediction of building-floor-location information as a classification task only and thus is expected to generate discretized, coarse-grained location estimates. The FedCRNP has been trained on not only the same training data partitions used by us but also the original testing data not used by us, and the latter part has been pseudo-labeled using a pretrained model before being evenly distributed to the clients to take advantage of this approach's capability of semi-supervised learning. The model architecture of the FedCRNP has not been changed, while some hyperparameters of it such as batch size have been tuned to achieve better results in our own experimental settings.

The localization accuracy of our FedLoc3D framework has been verified, which is measured by the error distance (in meters) between predicted location and ground truth in the 3-D space. The overall training performance of the combinations of classification and regression approaches is compared in Fig. 18, when the average storey height is assumed to be 3 m so that the altitude any target is at can be estimated from the floor level it is on. It can be seen that for both the IID and non-IID cases, a combination of our FedCNN-BFC and FedADA-LLR converges to the best outcomes, and a combination of our FedDSC-BFC and FedADA-LLR does not incur much performance loss. In comparison, the integration of the

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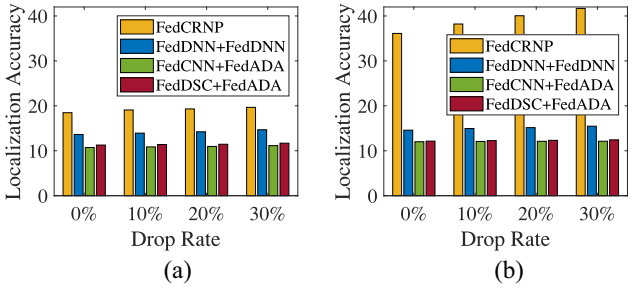

Fig. 19. Localization accuracy for FedLoc3D. (a) IID data. (b) Non-IID data.

FedDNN-BFC and FedDNN-LLR yields a relatively inferior alternative without enhanced feature extraction and generalization. Our implementation of the FedCRNP always produces the worst outcomes due to the inaccuracy of discretized LLR, and it even does not converge well in the worst-case non-IID setting. Without simplifying the issue of LLR to a classification task, our FedLoc3D can generally deliver at least a 40% performance gain over the FedCRNP. The average accuracy performance of the combination approaches is compared in Fig. 19, and similar results can be observed. The combinations of our approaches significantly outperform that of the benchmarks in such a typical multibuilding and multifloor environment. Under the impacts of data heterogeneity and network instability, our approaches not only push toward greater localization accuracy but also achieve even higher learning efficiency as indicated in the above two sections.

VII. Conclusion

It is challenging to jointly handle the issues of BFC and LLR in a sizable 3-D indoor space through enabling FL on decentralized yet heterogeneous data and over an imperfect wireless network. In this article, we confront these challenges and have proposed an FL framework, FedLoc3D, for both BFC and LLR. On the one hand, our FedDSC-BFC approach addresses the former issue by generating a multilabel classification model based on a 1-D CNN with depthwise separable convolutions. On the other hand, our FedADA-LLR approach addresses the latter issue by developing a multitarget regression model based on a DNN with autoencoder and data augmentation. Our approaches have been validated by extensive experiments on a real-world data set of WiFi fingerprints to improve both localization accuracy and learning efficiency under data heterogeneity and network instability. The new designs of our approaches can bring great benefits in feature extraction and generalization, and thus enhance the accuracy performance. Although the model sizes of our approaches rise to a certain extent, the overhead performance is not harmed but is even improved thanks to significant gains in training convergence.

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